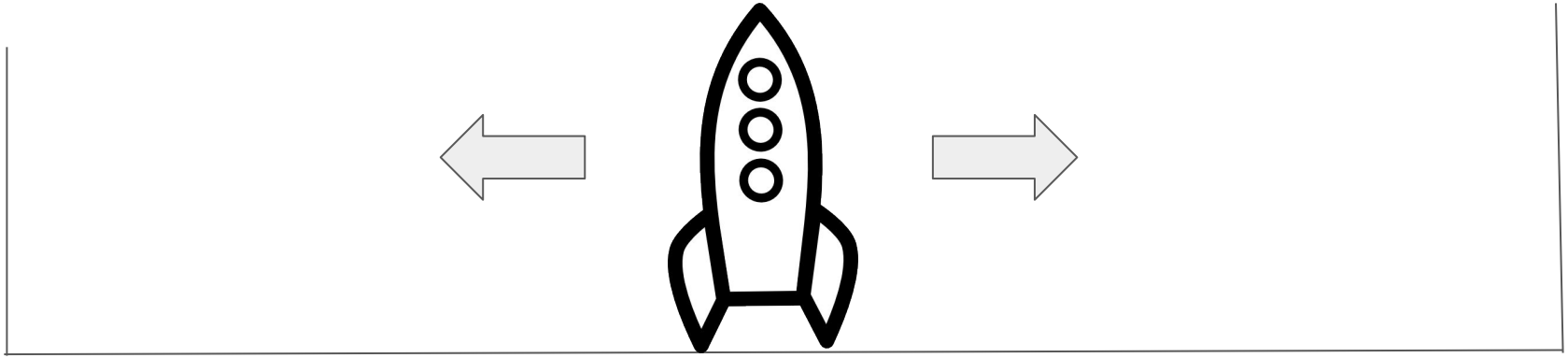
What happens when user presses left/right keys on keyboard?

- The ship only moves horizontally, so only the horizontal coordinate value of ship gets affected
- Decreases when left key is pressed, increases for right key
- Make sure the ship doesn’t go outside the window while you move it
- Make use of window width and ship width smartly to tackle this

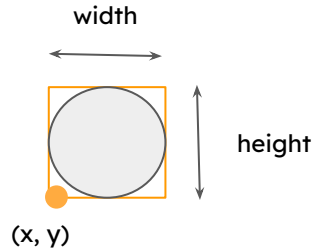


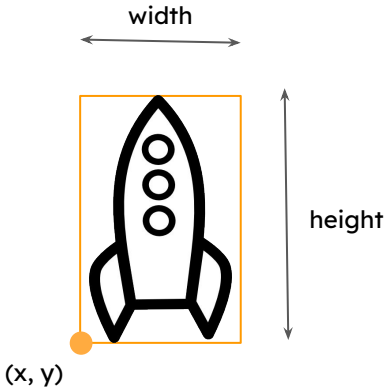
How can we ensure the ship/enemy or bullet/enemy when they come into contact?

- Imagine the objects have their respective axis-aligned bounding boxes (AABB) or Hitbox.
- AABB of an object has the same coordinates of the object itself (controlpoint), so they move together.
- Four properties (x, y, width, height). Width and Height are constant.
- We can check for collision/intersection of these two AABBs.

```
class AABB {
    public int x;
    public int y;
    public int width;
    public int height;
}
```

If you consider AABB/Hitbox control point to be at the bottom-left corner of an object

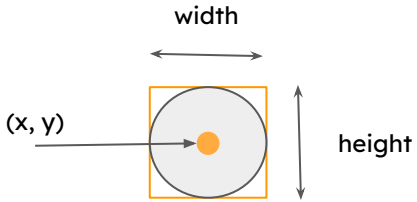


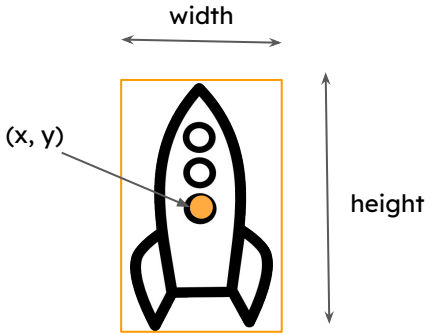



```
bool hasCollided(AABB box1, AABB box2) {
    return (box1.x) < (box2.x + box2.width)  &&
           (box1.x + box1.width) > (box2.x)  &&
           (box1.y) < (box2.y + box2.height) &&
           (box1.y + box1.height) > (box2.y);
}

// basically, we are checking these 4 conditions
// box1's xmin is leftwards  of box2's xmax
// box1's xmax is rightwards of box2's xmin
// box1's ymin is below box2's ymax
// box1's ymax is above box2's ymin
```

If you consider AABB/Hitbox control point to be at the center of an object






```
bool hasCollided(AABB box1, AABB box2) {
    return (box1.x - box1.width/2) < (box2.x + box2.width/2)    &&
           (box1.x + box1.width/2) > (box2.x - box2.width/2)    &&
           (box1.y - box1.height/2) < (box2.y + box2.height/2)  &&
           (box1.y + box1.height/2) > (box2.y - box2.height/2);
}

// basically, we are checking these 4 conditions
// box1's xmin is leftwards  of box2's xmax
// box1's xmax is rightwards of box2's xmin
// box1's ymin is below box2's ymax
// box1's ymax is above box2's ymin
```

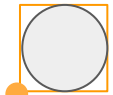
Calling Strategy

We call the hasCollided function in our animate/idle function
as many times as needed

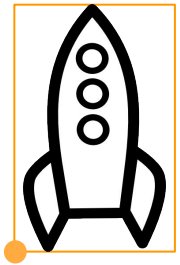
```
// detecting a successful kill or hit
bool hit_success = hasCollided(enemy_aabb, bullet_aabb);

// detecting a crash
bool crashed = hasCollided(enemy_aabb, rocket_aabb);
```

Collision Example



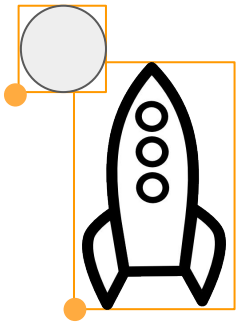
(10, 80)



(30, 5)

No collision

(10, 30)



(30, 5)

Collided

How to detect if an enemy sinks below window?

Monitor the enemy's coordinate value

And what happens if an enemy sinks?

Consult the question, you already know that



How to detect click over on-screen buttons?

- Buttons also have their own AABBs



- Calculate axial bounds:

```
xmin = x
ymin = y
xmax = x + width
ymax = y + height
```

- A successful click is registered when the mouse coordinates fall in the axial ranges:

$$xmin \leq mx \leq xmax$$
$$\text{and}$$
$$ymin \leq my \leq ymax$$

(mx, my) are converted (y-flipped) coordinates from mouse input. Obtained from mouse listener

